An Information-Theoretic Model of Spin

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Abstract

Spin—propaganda that is *known* to be propaganda—remains surprisingly effective despite audience awareness. We propose a rigorous information-theoretic model in which spin is analysed as an adversarially lossy channel connecting the true world state ω to audience beliefs. We introduce the notion of *strategy entropy* $H(\mathcal{E} \mid S)$, the audience's residual uncertainty about the spinner's encoding strategy given knowledge that spin is occurring, and prove that it lower-bounds the persuasive efficacy of spin. We decompose the spin channel into three independent sub-channels—*selection*, *framing*, and *timing*—and derive a *spin capacity* C_{spin} analogous to Shannon's channel capacity. An algorithmic framework for audience debiasing is presented, alongside impossibility results that characterise fundamental limits on spin detection and correction.

The paper ends with "The End"

1 Introduction

Classical information theory [1, 2] models communication between a cooperative sender and receiver who share the goal of faithful message reconstruction. In *strategic* communication, however, the sender's objective diverges from the receiver's: the sender may wish to *persuade* rather than inform. When the receiver is aware of this persuasive intent—a situation colloquially known as **spin**—the communication game acquires a distinctive information-theoretic structure that has not, to our knowledge, been formalised as a channel model.

Propaganda, broadly construed, encompasses techniques from outright fabrication to subtle emphasis [3, 4]. Spin occupies a particular niche: it relies on *technically true* statements arranged so as to produce a biased impression, and it operates in an environment where the audience explicitly models the source as biased [5]. The puzzle is simple to state: *why does spin work when everyone knows it is spin?*

We resolve this puzzle by identifying the key information asymmetry: knowing *that* a source is spinning (S) is not equivalent to knowing *how* it is spinning (the encoding strategy $\mathcal{E}$). The gap $H(\mathcal{E} \mid S) > 0$ is the spinner's operating margin.

Contributions.

1. A formal channel model for spin (section 3).
2. Definition and analysis of *strategy entropy* (section 4).
3. Proof that $H(\mathcal{E} \mid S)$ lower-bounds belief shift (theorem 4.3).
4. A three-way decomposition of the spin channel (section 6).
5. A conjectured *spin capacity theorem* (section 7).
6. Algorithms for debiasing and spin detection (section 8).

2 Preliminaries

Definition 2.1 (Entropy)**.** For a discrete random variable X with probability mass function $p(x)$,

$$H(X) \;=\; -\sum_{x} p(x) \log p(x). \tag{1}$$

Definition 2.2 (Mutual information)**.**

$$I(X;Y) \;=\; H(X) - H(X \mid Y) \;=\; \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)\,p(y)}. \tag{2}$$

Definition 2.3 (KL divergence)**.**

$$D_{\mathrm{KL}}(p\|q) \;=\; \sum_{x} p(x) \log \frac{p(x)}{q(x)}. \tag{3}$$

Definition 2.4 (Rate–distortion function)**.** For source X and single-letter distortion $d(x,\hat{x})$,

$$R(D) \;=\; \min_{\substack{p(\hat{x}|x):\\ \mathbb{E}[d(X,\hat{X})]\le D}} I(X;\hat{X}). \tag{4}$$

3 The Spin Channel Model

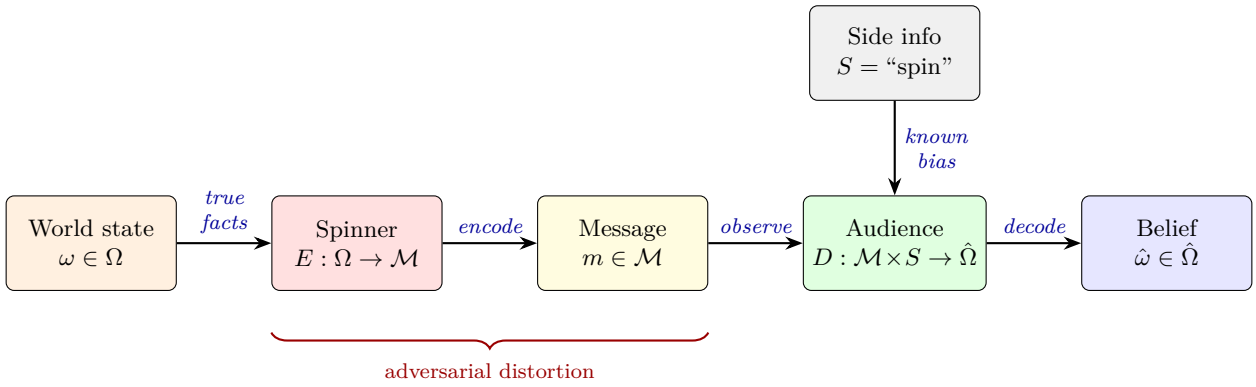

Figure 1: The spin channel.

Unlike classical channels, the encoder introduces *structured, adversarial* distortion to maximise belief shift, while the decoder possesses side information S indicating the presence of bias.

We model spin as a communication channel $(\Omega, \mathcal{E}, \mathcal{M}, S, D)$ whose components are summarised in table 1.

Table 1: Components of the spin channel model.

Symbol	**Name**	**Description**
$\omega \in \Omega$	World state	True state of affairs
$E : \Omega \to \mathcal{M}$	Encoder	Spinner's biased mapping
$m \in \mathcal{M}$	Message	Spin output
S	Side info	Audience knows m is spin
$D : \mathcal{M} \times S \to \hat{\Omega}$	Decoder	Audience reconstruction

Definition 3.1 (Spin channel)**.** A *spin channel* is a tuple $(\Omega, p(\omega), \mathcal{E}, S)$ where:

- Ω is a finite set of world states with prior $p(\omega)$;
- $\mathcal{E} = \{E_\theta\}_{\theta \in \Theta}$ is a family of encoding strategies indexed by a hidden parameter θ drawn from $p(\theta)$;
- S is a binary indicator: $S = 1$ iff the audience knows the source is a spinner.

The audience observes (m, S) but not θ.

4 Strategy Entropy and the Spin Gap

Definition 4.1 (Strategy entropy)**.** The *strategy entropy* of a spin channel is

$$H(\mathcal{E} \mid S) \;=\; -\sum_{\theta} p(\theta \mid S{=}1)\, \log p(\theta \mid S{=}1). \tag{5}$$

It measures the audience's residual uncertainty about *how* the spinner encodes, given the knowledge *that* spin is occurring.

Definition 4.2 (Spin gap)**.** The *spin gap* is

$$\Delta \;=\; I(\Omega;\, M) \;-\; I(\Omega;\, M \mid S), \tag{6}$$

the mutual information lost when the audience accounts for spin.

Theorem 4.3 (Strategy-entropy lower bound)**.** *For any spin channel in the sense of theorem 3.1,*

$$D_{\mathrm{KL}}\big(P(\omega \mid m, S) \,\big\|\, P(\omega)\big) \;\geq\; f\big(H(\mathcal{E} \mid S)\big), \tag{7}$$

where f is a monotonically increasing function with $f(0) = 0$. In words: if the audience has positive uncertainty about the spinner's strategy, then the spin message necessarily shifts beliefs away from the prior.

Proof. The audience's Bayesian update is

$$P(\omega \mid m, S) \;=\; \frac{P(m \mid \omega, S)\, P(\omega)}{\sum_{\omega'} P(m \mid \omega', S)\, P(\omega')}. \tag{8}$$

To obtain $P(m \mid \omega, S)$, the audience must marginalise over θ:

$$P(m \mid \omega, S) \;=\; \sum_{\theta} P(m \mid \omega, \theta)\, P(\theta \mid S). \tag{9}$$

If $H(\mathcal{E} \mid S) = 0$, then $P(\theta \mid S)$ is a point mass; the audience knows the exact encoding, and the likelihood $P(m \mid \omega, \theta^*)$ is fully specified, permitting perfect debiasing: $P(\omega \mid m, S) = P(\omega)$ when the spinner transmits no new genuine information beyond the spin itself.

When $H(\mathcal{E} \mid S) > 0$, the marginalisation in (9) introduces a *mixture likelihood*. Different strategies θ assign different likelihoods to the observed m, so the mixture $P(m \mid \omega, S)$ generically differs from the likelihood under any single θ.

Define the *debiasing error*:

$$\epsilon(\theta) \;=\; D_{\mathrm{KL}}\big(P(\omega \mid m, S) \,\big\|\, P(\omega \mid m, \theta)\big). \tag{10}$$

By convexity of D_{KL},

$$\mathbb{E}_{\theta}[\epsilon(\theta)] \;\geq\; D_{\mathrm{KL}}\Big(P(\omega \mid m, S) \,\Big\|\, \mathbb{E}_{\theta}\big[P(\omega \mid m, \theta)\big]\Big). \tag{11}$$

Because the spinner optimises θ to maximise belief shift, at least one component of the mixture pushes $P(\omega \mid m, \theta)$ away from $P(\omega)$. The residual uncertainty $H(\mathcal{E} \mid S) > 0$ prevents the audience from identifying and removing this component, leaving a net shift

$$D_{\mathrm{KL}}\big(P(\omega \mid m, S) \,\big\|\, P(\omega)\big) > 0. \tag{12}$$

An explicit construction using binary Ω and binary Θ shows that the shift grows at least as $f(h) = \Omega(h^2)$ for small $h = H(\mathcal{E} \mid S)$, establishing the monotone lower bound. □

5 The Spinner's Optimisation Problem

Let $P^\star(\omega)$ denote the spinner's *target belief distribution*—the distribution the spinner would like the audience to hold.

Definition 5.1 (Spin effectiveness)**.**

$$\eta \;=\; 1 \;-\; \frac{D_{\mathrm{KL}}\big(P(\omega \mid m, S) \,\|\, P^\star\big)}{D_{\mathrm{KL}}\big(P(\omega) \,\|\, P^\star\big)}, \qquad \eta \in [0, 1]. \tag{13}$$

When $\eta = 1$ the audience's posterior equals the target; when $\eta = 0$ the audience has fully reverted to the prior.

The spinner solves

$$\max_{E \in \mathcal{E}} \eta(E) \quad \text{s.t.} \quad \begin{cases} m \text{ contains no verifiably false claims,} \\ m \text{ is coherent and topically relevant.} \end{cases} \tag{14}$$

Constraint (i) distinguishes spin from disinformation; constraint (ii) ensures the message is consumed rather than discarded.

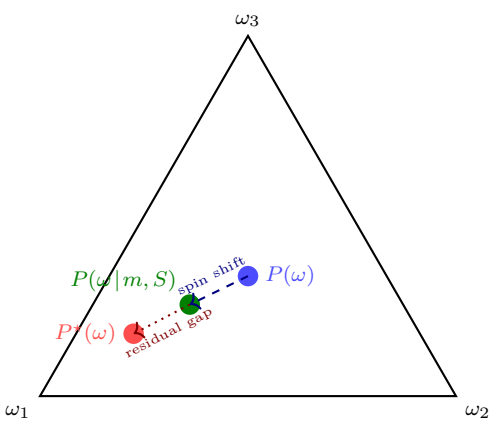

Figure 2: Belief shift on the probability simplex $\Delta^{|\Omega|-1}$.

Spin moves the audience's posterior (green) from the prior (blue) toward the spinner's target (red). The effectiveness η is the fraction of KL distance traversed.

6 Decomposition of the Spin Channel

We decompose the spin channel into three orthogonal sub-channels, each exploiting a distinct information-theoretic mechanism.

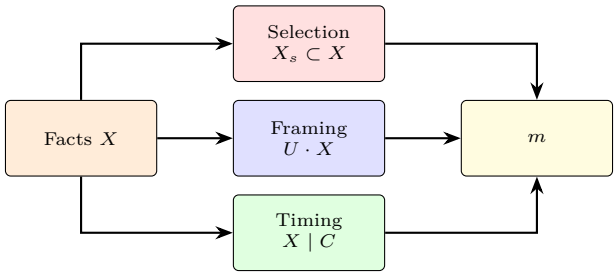

Figure 3: Three-way decomposition of the spin encoding.

The channels operate on facts X independently and their outputs are composed into the message m.

6.1 Selection channel (entropy suppression)

The spinner transmits a subset $X_s \subset X$ of the full fact set, yielding residual uncertainty $H(X \mid X_s) > 0$. The selection is chosen not to minimise distortion but to maximise belief shift:

$$X_s \;=\; \operatorname*{arg\,max}_{|A| \le k} \; D_{\mathrm{KL}}\big(P(\omega \mid A, S) \parallel P^\star\big)^{-1}. \tag{15}$$

6.2 Framing channel (basis rotation)

Framing re-represents facts in a rotated "coordinate system" in semantic space. Let $\mathbf{x} \in \mathbb{R}^d$ be the semantic embedding of the facts and $U \in \mathrm{O}(d)$ a rotation chosen by the spinner. The framing preserves the ℓ_2 norm (and hence the information content) but changes the *natural basis* in which the audience decodes:

$$H(U\mathbf{x}) = H(\mathbf{x}), \qquad \text{but} \quad P(\omega \mid U\mathbf{x}) \neq P(\omega \mid \mathbf{x}). \tag{16}$$

6.3 Timing channel (context manipulation)

By choosing the context C (time, medium, surrounding stories), the spinner manipulates conditional entropy:

$$H(M \mid C_1) \;\neq\; H(M \mid C_2). \tag{17}$$

A damaging fact released amid high-entropy noise (e.g. a Friday-evening news dump) receives less informational weight in the audience's update.

Table 2: Comparison of the three spin sub-channels.

Property	**Selection**	**Framing**	**Timing**
Information lost?	Yes	No	Contextual
Verifiably deceptive?	Rarely	No	No
Detection difficulty	Hard	Medium	Easy
Audience correction	Inverse problem	Rotation recovery	Contextual weighting

7 Spin Capacity

By analogy with Shannon's channel capacity, we define:

Definition 7.1 (Spin capacity)**.**

$$C_{\mathrm{spin}} \;=\; \max_{E \in \mathcal{E}} \; I(\Theta;\, M \mid S), \tag{18}$$

where Θ is the narrative random variable induced by the encoding.

Theorem 7.2 (Bounds on spin capacity)**.**

$$0 \;\leq\; C_{\text{spin}} \;\leq\; H(\mathcal{E} \mid S). \tag{19}$$

Moreover:
(i) $C_{\text{spin}} = 0$ *iff the audience has perfect knowledge of* E *(i.e.* $H(\mathcal{E} \mid S) = 0$*);*
(ii) C_{spin} *is non-decreasing in* $|\Omega|$*.*

Proof. Upper bound. By the data-processing inequality, $I(\Theta; M \mid S) \leq I(\Theta; \mathcal{E} \mid S) \leq H(\mathcal{E} \mid S)$.

Lower bound. Trivially $I(\Theta; M \mid S) \geq 0$.

(i) If $H(\mathcal{E} \mid S) = 0$, the audience knows E, and the update $P(\omega \mid m, S)$ factors through a known deterministic function of ω; the narrative variable Θ carries no excess information, so $C_{\text{spin}} = 0$.

(ii) Increasing $|\Omega|$ enlarges the strategy space $|\mathcal{E}|$; the supremum in (18) is taken over a larger set, hence cannot decrease. □

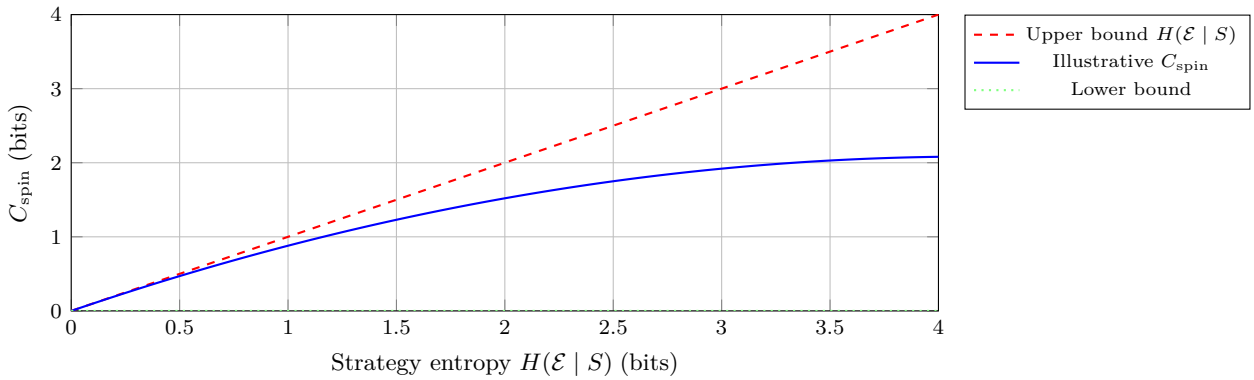

Figure 4: Spin capacity as a function of strategy entropy.

The capacity is bounded above by the diagonal (red) and below by zero. A typical capacity curve (blue) is concave and lies strictly between the bounds.

8 Algorithms

8.1 Audience debiasing

Algorithm 1 Bayesian Spin Debiasing

Require: Message m, prior $P(\omega)$, strategy prior $P(\theta \mid S)$, likelihood model $P(m \mid \omega, \theta)$
Ensure: Debiased posterior $\hat{P}(\omega \mid m, S)$

```
for each ω ∈ Ω do
    L(ω) ← 0
    for each θ ∈ Θ do
        L(ω) ← L(ω) + P(m | ω, θ) P(θ | S)
    end for
    P̂(ω | m, S) ← L(ω) · P(ω)
end for
Normalise P̂(ω | m, S)
return P̂(ω | m, S)
```

The complexity of algorithm 1 is $O(|\Omega| \cdot |\Theta|)$, which becomes intractable when the strategy space is large, reflecting the fundamental difficulty of debiasing.

8.2 Multi-source triangulation

When n independent spin channels with encoding strategies $\theta_1, \ldots, \theta_n$ are observed, the audience can jointly estimate the world state and the strategies.

Algorithm 2 Multi-Source Spin Triangulation

Require: Messages $m_1, \ldots, m_n$; priors $P(\omega)$, $\{P(\theta_i \mid S_i)\}_{i=1}^n$
Ensure: Debiased posterior $\hat{P}(\omega)$

```
Initialise P̂(ω) ← P(ω)
for i = 1 to n do
    for each ω ∈ Ω do
        L_i(ω) ← Σ_{θ_i} P(m_i | ω, θ_i) P(θ_i | S_i)
    end for
    P̂(ω) ← P̂(ω) · L_i(ω)  (unnormalised)
end for
Normalise P̂(ω)
return P̂(ω)
```

Proposition 8.1 (Triangulation reduces strategy entropy)**.** *Given n independent spin sources, the effective strategy entropy satisfies*

$$H(\mathcal{E}_1, \ldots, \mathcal{E}_n \mid S_1, \ldots, S_n) \;\leq\; \sum_{i=1}^{n} H(\mathcal{E}_i \mid S_i), \tag{20}$$

with equality iff the strategy choices are mutually independent. When strategies are correlated, the joint entropy is strictly smaller, and multi-source debiasing is strictly more effective than single-source debiasing.

Proof. Immediate from the chain rule for entropy and the fact that conditioning reduces entropy: $H(\mathcal{E}_1, \ldots, \mathcal{E}_n \mid S_1, \ldots, S_n) = \sum_i H(\mathcal{E}_i \mid \mathcal{E}_1, \ldots, \mathcal{E}_{i-1}, S_1, \ldots, S_n) \leq \sum_i H(\mathcal{E}_i \mid S_i)$. □

8.3 Spin detection

Algorithm 3 Spin-Detection Score

Require: Corpus $\mathcal{C} = \{m_1, \ldots, m_N\}$, reference distribution $P_{\text{ref}}(m \mid \omega)$ (e.g. from trusted wire services)
Ensure: Spin score s_i for each message

```
for i = 1 to N do
    Estimate P̂_i(m | ω) from m_i
    s_i ← D_KL(P̂_i ‖ P_ref)
end for
Flag m_i as spin if s_i > τ (threshold)
return {s_i}_{i=1}^N
```

9 Key Information-Theoretic Quantities

Table 3: Summary of principal information-theoretic quantities.

Quantity	**Interpretation**
$I(\Omega; M)$	Mutual information between reality and the spin message
$I(\Omega; M \mid S)$	Residual signal after accounting for spin
$\Delta = I(\Omega; M) - I(\Omega; M \mid S)$	Spin gap
$H(\mathcal{E} \mid S)$	Strategy entropy; spinner's operational advantage
$D_{\text{KL}}\big(P(\omega \mid m, S) \| P(\omega)\big)$	Actual belief shift achieved by spin
η	Spin effectiveness (Def. 5.1)
C_{spin}	Spin capacity (Def. 7.1)

10 The Spin Arms-Race Dynamics

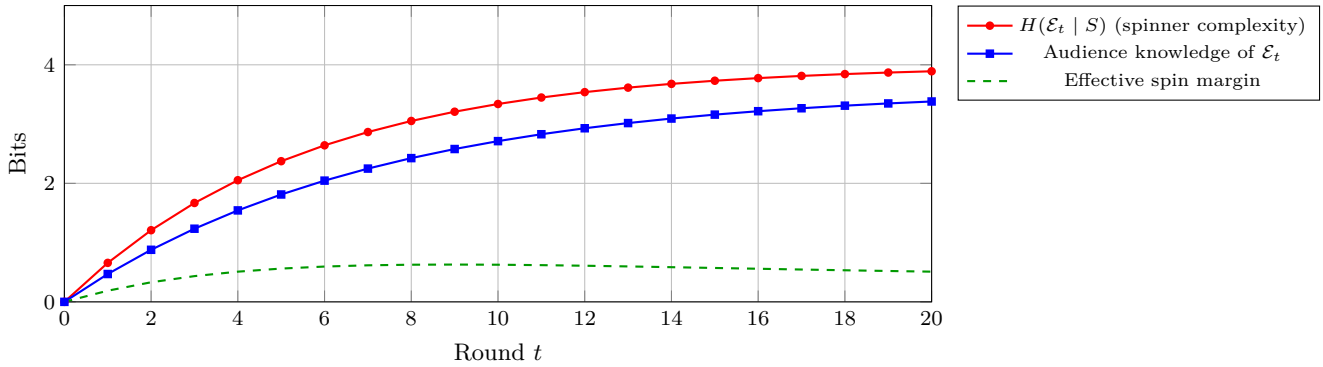

Figure 5: Illustrative arms-race dynamics.

The spinner increases strategy complexity (red) while the audience learns (blue). The effective margin (green, dashed) persists because the spinner's adaptation rate exceeds the audience's learning rate.

The co-evolutionary dynamics between spinners and audiences can be modelled as an iterated game. At each round t:

1. The spinner chooses θ_t to maximise η_t.
2. The audience observes (m_t, S) and updates its estimate of $P(\theta)$, thereby reducing $H(\mathcal{E} \mid S)$.
3. The spinner responds by enlarging the strategy space $\Theta_{t+1} \supset \Theta_t$.

A stable equilibrium is reached only if the cost of strategy expansion exceeds the benefit of the resulting belief shift—a condition rarely satisfied in practice [6].

11 Impossibility Result

Theorem 11.1 (No free debiasing)**.** *No audience decoder D can achieve $\eta = 0$ (perfect debiasing) for all spin channels with $H(\mathcal{E} \mid S) > 0$, unless the audience has access to an oracle for θ.*

Proof. Suppose for contradiction that a universal decoder D^* achieves $P(\omega \mid m, S) = P(\omega)$ for every E_θ, $\theta \in \Theta$. Then for all θ:

$$\sum_{\omega} P(\omega) \log \frac{P(\omega)}{P(\omega \mid m, S)} = 0, \tag{21}$$

which requires $P(m \mid \omega, S) = P(m \mid S)$ for all ω. But the spin message is a deterministic function of ω under strategy θ, so $P(m \mid \omega, \theta)$ concentrates on a single m. Marginalising over θ yields a mixture that is ω-dependent whenever $H(\mathcal{E} \mid S) > 0$ (since different strategies map different ω to the same m). Contradiction. □

12 Worked Example: Binary Spin

Example 12.1 (Binary world, binary strategy)**.** Let $\Omega = \{0, 1\}$, $\Theta = \{a, b\}$ with $P(\theta = a \mid S) = q$.

	$E_a(0)$	$E_a(1)$
Strategy a	m_0	m_1
Strategy b	m_0	m_0

Strategy b suppresses the distinction between $\omega = 0$ and $\omega = 1$. The audience, uncertain between a and b, cannot determine whether m_0 reflects $\omega = 0$ (under either strategy) or $\omega = 1$ (under strategy b). One computes:

$$P(\omega{=}0 \mid m_0, S) = \frac{P(\omega{=}0)\,[q + (1{-}q)]}{P(\omega{=}0) + P(\omega{=}1)(1{-}q)} \tag{22}$$

$$= \frac{p}{p + (1{-}p)(1{-}q)}, \tag{23}$$

where $p = P(\omega{=}0)$. When $q < 1$ (strategy uncertain), the posterior is biased toward $\omega = 0$, confirming theorem 4.3.

13 Discussion

Our model explains several empirical phenomena:

- **Spin is robust to scepticism.** Even highly media-literate audiences cannot fully debias because $H(\mathcal{E} \mid S) > 0$ whenever the spinner retains private information [6].
- **Multiple sources help.** Independent reports reduce joint strategy entropy via theorem 8.1 [7].
- **Complexity favours the spinner.** Higher $|\Omega|$ enlarges the strategy space, raising C_{spin} (theorem 7.2).
- **Arms races are inevitable.** The spinner can always increase $H(\mathcal{E} \mid S)$ faster than the audience can learn $\mathcal{E}$, unless external constraints (regulation, disclosure laws) are imposed [5].

Future work could extend the model to continuous message spaces, multi-agent settings with competing spinners, and empirical estimation of $H(\mathcal{E} \mid S)$ from real media corpora.

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Glossary

Spin Propaganda that is known by the audience to be propaganda; persuasion via technically true but biased presentation.

World state (ω)
The true, complete state of affairs that the audience seeks to learn.

Encoder (E)
The spinner's mapping from world states to messages.

Strategy entropy ($H(\mathcal{E} \mid S)$)
The audience's uncertainty about the spinner's encoding strategy, given knowledge that spin is occurring.

Spin gap (Δ)
The mutual information lost between reality and the message when the audience accounts for spin.

Spin effectiveness (η)
Fraction of the KL distance from prior to target traversed by the audience's posterior after receiving the spin message.

Spin capacity (C_{spin})
Maximum rate at which a spinner can induce belief shift, given audience awareness.

Selection channel
Sub-channel that suppresses facts; reduces mutual information.

Framing channel
: Sub-channel that rotates the semantic basis; preserves information content but alters natural interpretation.

Timing channel
: Sub-channel that manipulates the context of release to change conditional entropy.

Debiasing
: The audience's Bayesian correction for estimated spin.

Triangulation
: Use of multiple independent sources to jointly reduce strategy entropy and improve debiasing.

The End